

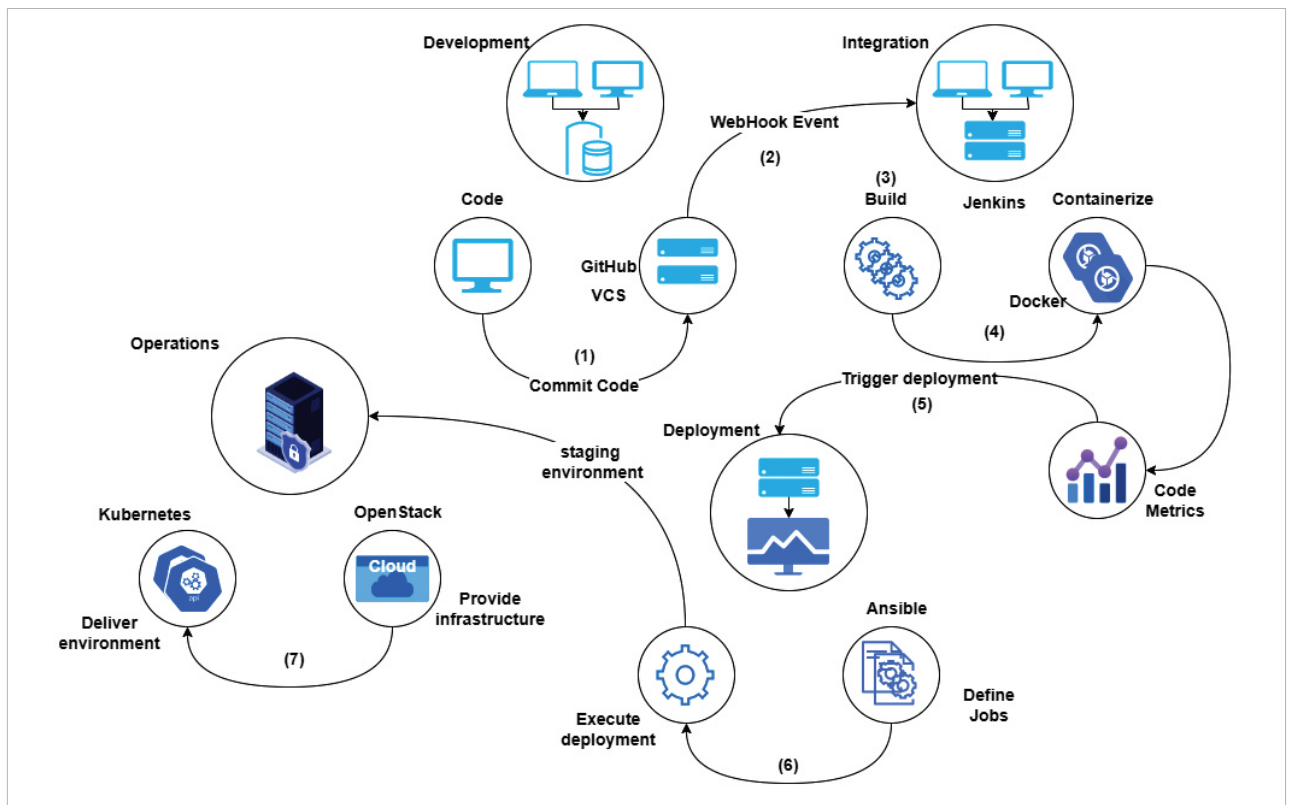


Figure 9: NFV integration with DevOps pipeline

experimental setup and can be rolled back if and when needed. A deployment infrastructure can be structured within no time with the help of OpenStack which introduces abstract computation, networking, and storage services through its various components. With OpenStack, the administrator can feed new virtual machines to the Kubernetes cluster so that it can stay on par with the growing traffic. This project can further be extended to various NFV use cases such as content delivery networks (CDNs), load balancers, web application firewalls, and intrusion detection and prevention systems.

This DevOps integration in network functions virtualization will help scale and readily deploy the applications. Microservices-based architecture is already ruling NFV deployments. When combined with DevOps, the automated pipeline will enable independent and dynamic scaling of the functional elements of the NFV system. This will

help reduce the capex and opex of the NFV market. All these tool-chains provide a collective development and delivery solution, which not only reduces the time of deployment but also the workload on developers, testers, and operators. The entire pipeline will give any NFV module deliverable an edge over traditional deployments.

In this comprehensive eight-article series, we have extensively covered essential topics in today's technology landscape. We explored distributed systems, containerisation, Kubernetes, open source cloud computing, software-defined networking (SDN), network functions virtualization (NFV), DevOps methodologies, and the integration of network functions virtualization infrastructure (NFVI). By delving

into these subjects, we provided a comprehensive understanding of how these technologies and concepts can be seamlessly integrated for optimal performance in modern IT environments. We demonstrated the practical implementation of NFV integration with the DevOps pipeline, showcasing its potential to enhance network operations and accelerate software delivery.

We believe that this series has equipped you with valuable insights and actionable knowledge to embark on your own successful journey in implementing these powerful solutions. May your endeavours in embracing NFV and DevOps be met with efficiency, innovation, and success in the ever-evolving world of technology. **END** 🐧

By: Shubham Aggarwal, Nithya Ganesan, and B. Thangaraju

The authors are associated with the Open Source Technology Lab at the International Institute of Information Technology, Bengaluru.